Presentation Script - Friction Modelling of a 6-DOF Robotic Arm

Slide 1: Title Slide

Good morning everyone. I am Soumyadeep Dey from the National Institute of Technology Durgapur. Today I will present my internship project carried out at CSIR-CMERI on Friction Modelling of a 6-DOF Robotic Arm Using Physics-Based and Data-Driven Approaches. The objective of this work is to understand friction behaviour in robotic joints and compare classical friction models with machine learning approaches using real experimental data.

Slide 2: Introduction & Background

Industrial robotic manipulators contain friction due to gearboxes, bearings, harmonic drives, and elastic joint components. These friction effects reduce positioning accuracy and affect trajectory tracking performance. Although modern controllers compensate for friction, analysis of the experimental data showed that residual friction effects still exist. Therefore, accurate friction modelling is important for improving robot performance and achieving precise motion control.

Slide 3: Project Objectives

The main objectives of this project are to study the friction characteristics of the 6-DOF robotic arm developed by Svaya Robotics, compare classical physics-based friction models with machine learning models, and investigate additional dynamic effects observed in the data, particularly the phase difference between motor torque and joint torque signals.

Slide 4: Problem Statement

The robotic arm provides measurements of motor torque, joint torque, joint velocity, and joint position. Using these signals, we define a residual torque term: Tf equals N times motor torque minus joint torque, where N is the gearbox ratio of 100. This residual torque is not pure friction and also contains transmission losses, gearbox compliance effects, elastic dynamics, and other unmodelled behaviours.

Slide 5: Platform & Data Collection

The experimental platform is a six-degree-of-freedom robotic arm equipped with harmonic drive transmissions. Data was collected at 200 Hz and approximately 45,000 samples were recorded. Forward and reverse motions were executed to capture velocity reversals and nonlinear friction characteristics.

Slide 6: Literature Review & Research Gap

Several friction models including Coulomb, Viscous, Stribeck, LuGre, Dahl, and General Maxwell Slip models were studied. Most previous research relies heavily on simulation environments and often neglects real-world environmental disturbances and robot-specific dynamics. This project addresses that gap using real experimental data.

Slide 7: Physics-Based Friction Modelling

The first approach is classical friction modelling using the Coulomb plus viscous model and the Stribeck model. These models capture major friction characteristics but residual analysis revealed additional nonlinear dynamics that are not fully represented.

Slide 8: Physics-Based Results

Using random train-test splitting and unique data points, the final R-squared values obtained were 0.9782 for Joint 1, 0.7705 for Joint 2, 0.7262 for Joint 3, and 0.9326 for Joint 5. These results show that physics-based models capture the main friction trends but struggle with complex nonlinear effects.

Slide 9: Data-Driven Modelling

The second approach is a black-box neural network. The inputs are joint position and velocity, while the output is friction-related torque. The network learns directly from experimental data without requiring explicit friction assumptions.

Slide 10: Evaluation Metrics

Three metrics were used: RMSE, MAE, and R-squared score. RMSE and MAE measure prediction error, while R-squared measures how much variation in the data is explained by the model.

Slide 11: Neural Network Results

The most realistic test involved training on certain velocity ranges and testing on previously unseen velocities. The neural network achieved R-squared values of 0.9961, 0.8784, 0.9800, and 0.9906 for Joints 1, 2, 3, and 5 respectively.

Slide 12: Comparative Analysis

For every joint, the neural network achieved higher R-squared values than the classical friction model. This demonstrates the ability of machine learning to capture complex nonlinear friction behaviour present in real robotic systems.

Slide 13: Conclusions

Friction is a major contributor to torque losses in the robotic arm. The observed phase difference between motor torque and joint torque suggests spring-mass-damper dynamics caused by harmonic drive elasticity. Machine learning models provide significantly better predictive performance than simple physics-based models.

Slide 14: Future Work

Future work includes implementation of the LuGre dynamic friction model, development of Physics-Informed Neural Networks, and modelling of coupled multi-joint dynamics.

Slide 15: Thank You

Thank you for your attention. I would be happy to answer any questions.